10.1 Parametrizations of Plane Curves

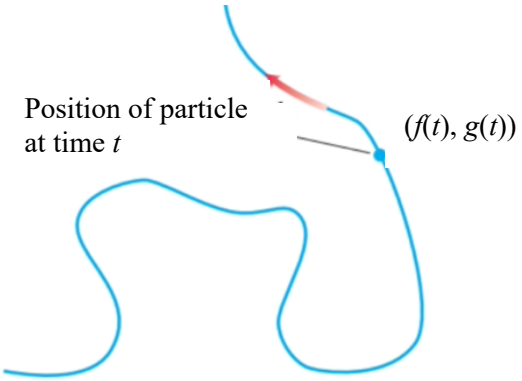


Figure 1 shows the path of a moving particle in the xy-plane. Notice that the path fails the vertical line test, so it cannot be described as the graph of a function of the variable x. But it can be described by a pair of equations

$$x = f(t),\ y = g(t)$$

where f and g are continuous functions.

Figure 1: The curve or path traced by a particle moving in the xy-plane is not always the graph of a function or single equation.

Definition 1: If x and y are given as functions

$$x = f(t),\ y = g(t)$$

over an interval I of t-values, then the set of points $(x, y) = (f(t), g(t))$ defined by these equations is a **parametric curve**. The equations are **parametric equations** for the curve.

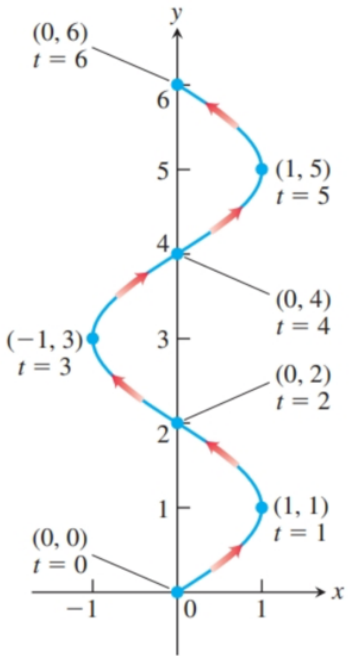
The variable t is a **parameter** for the curve, and its domain I is the **parameter interval**. If I is a closed interval, $a \le t \le b$, the point $(f(a), g(a))$ is the **initial point** of the curve and $(f(b), g(b))$ is the **terminal point**. When we give parametric equations and a parameter interval for a curve, we say that we have **parametrized** the curve. The equations and interval together constitute a **parametrization** of the curve. A given curve can be represented by different sets of parametric equations.

Example 1: Sketch the curve defined by the parametric equations

$$x = \sin \pi t/2,\ \ y = t,\ \ \ 0 \le t \le 6.$$

Solution:
Values of $x = \sin \pi t/2$ and $y = t$ for selected values of $0 \le t \le 6$.

t	x	y
0	0	0
1	1	1
2	0	2
3	−1	3
4	0	4
5	1	5
6	0	6



Make a table of values plot the points (x, y), and draw a smooth curve through them. If When $t = 0$, the initial point is (0, 0); when $t = 6$, the terminal point is (0, 6). The direction of motion is shown by the arrows on the curve.

Example 2: Sketch the curve defined by the parametric equations

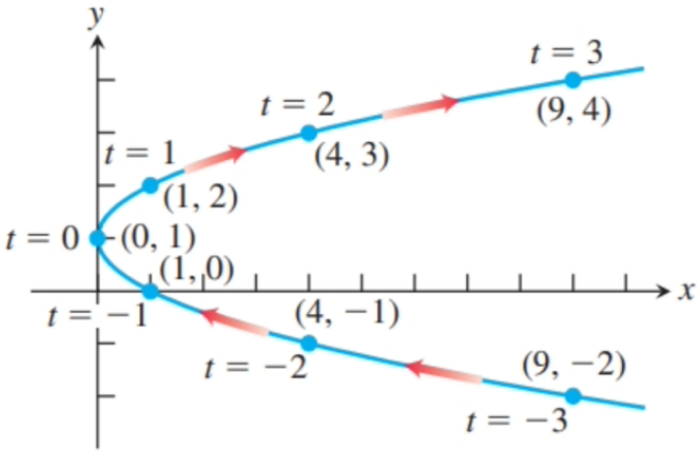
$$x = t^2,\ y = t + 1,\quad t \in \mathbb{R}.$$

Solution:

We make a table of values plot the points (x, y), and draw a smooth curve through them. If When $t = -3$, the point is $(9, -3)$; when $t = 3$, the point is $(9, 4)$. The direction of motion is shown by the arrows on the curve.

Values of $x = t^2$, and $y = t + 1$ for selected values of t.

t	x	y
−3	9	−2
−2	4	−1
−1	1	0
0	0	1
1	1	2
2	4	3
3	9	4



For this example we can use algebraic manipulation to eliminate the parameter t and obtain an algebraic equation for the curve in terms of x and y alone.

Substitute $t = y - 1$ into $x = t^2$, we have

$$x = t^2 = (y - 1)^2 = y^2 - 2y + 1.$$

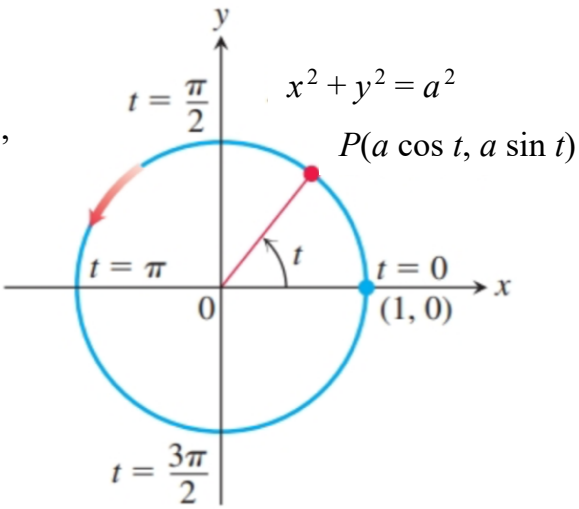
The equation $x = y^2 - 2y + 1$ represents a parabola, as displayed above.

Example 3: Graph the parametric curves

$$x = a\cos t,\ y = a\sin t,\quad 0 \le t \le 2\pi.$$

Solution:

Since $x^2 + y^2 = (a\cos t)^2 + (a\cos t)^2 = a^2$,
the parametric curve lies along the circle $x^2 + y^2 = a^2$,
centered at the origin with radius $r = |a|$.
As t increases from 0 to 2π,
the point $(x, y) = (a\cos t, a\sin t)$ starts at $(a, 0)$ and traces the entire circle once counterclockwise,
returning to $(a, 0)$ at $t = 2\pi$.



Example 4: The position $P(x, y)$ of a particle moving in the xy-plane is given by the equations and parameter interval

$$x = \sqrt{t} \quad , \quad y = t. \quad , \quad t \geq 0.$$

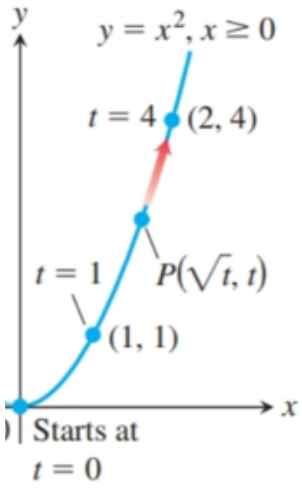
Identify the path traced by the particle and describe the motion.

Solution:

Eliminate t between the equations $x = t$ and $y = t$, we have hat

$$y = t = \left(\sqrt{t}\right)^2 = x^2 \quad , \quad x \geq 0.$$

Thus, the particle moves along the half of the parabola $y = x^2$, $x \geq 0$. The particle starts at (0, 0) when $t = 0$ and rises into the first quadrant as t increases. The parameter interval is $[0,\infty)$, and there is no terminal point.

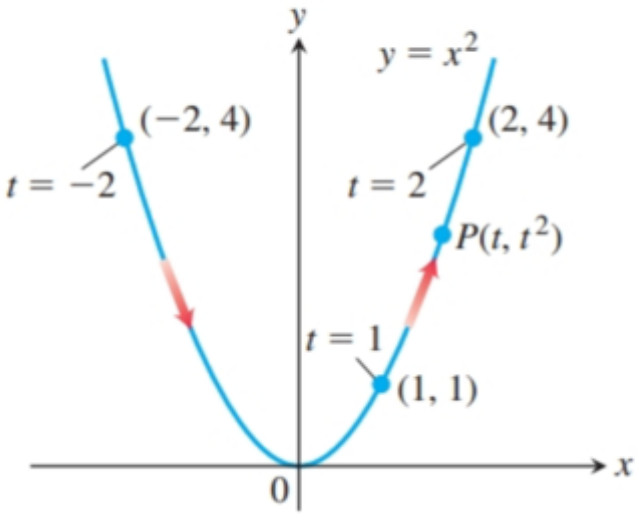


The graph of any function $y = f(x)$ can always be given a **natural parametrization** $x = t$ and $y = f(t)$. The domain of the parameter in this case is the same as the domain of the function f.

Example 5: A parametrization of the graph of the function $f(x) = x^2$ is given by

$$x = t \, , \quad y = f(t) = t^2 \quad , \quad t \in \mathbb{R}.$$

When $t \geq 0$, this parametrization gives the same path in the xy-plane as we had in Example 4. However, since the parameter t here can now also be negative, we have the entire parabolic curve. For this parametrization, there is no starting point and no terminal point.



Example 6: Find a parametrization for the line through the point (a, b) having slope m.

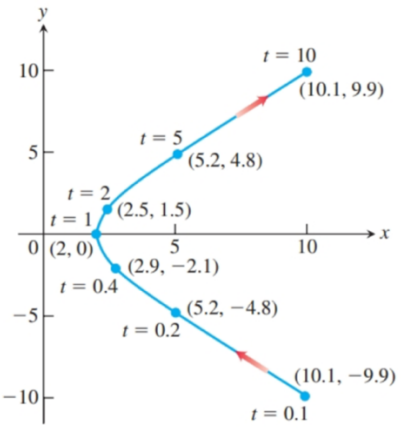
Solution:
A Cartesian equation of the line is $y - b = m(x - a)$. Define the parameter t by $t = x - a$, we have $x = a + t, \quad y = b + mt, \quad t \in \mathbb{R}$
parametrizes the line. This parametrization differs from the one we would obtain by the natural parametrization when $t = x$. However, both parametrizations describe the same line.

Example 7: Sketch and identify the path traced by the point $P(x, y)$ if

$$x = t + \frac{1}{t}, \quad y = t - \frac{1}{t}, \quad t > 0.$$

Solution:
Make a brief table of values, plot the points, and draw a smooth curve through them.
Values of $x = t + \frac{1}{t}, \quad y = t - \frac{1}{t}$,
for selected values of $t > 0$.

t	$1/t$	x	y
0.1	10.0	10.1	−9.9
0.2	5.0	5.2	−4.8
0.4	2.5	2.9	−2.1
1.0	1.0	2.0	0.0
2.0	0.5	2.5	1.5
5.0	0.2	5.2	4.8
10.0	0.1	10.1	9.9



$$x - y = \left(t + \frac{1}{t}\right) - \left(t - \frac{1}{t}\right) = \frac{2}{t} \qquad \text{---------- (1)}$$

$$x + y = \left(t + \frac{1}{t}\right) + \left(t - \frac{1}{t}\right) = 2t \qquad \text{---------- (2)}$$

Eliminate the parameter t from the equations (1) and (2), we have

$$(x - y)(x + y) = \left(\frac{2}{t}\right)(2t) = 4 \quad \Rightarrow \quad x^2 - y^2 = 4, \quad x > 0. \qquad \text{----- (3)}$$

Thus, the coordinates of all the points $P(x, y)$ described by the parametric equations satisfy Equation (3), which is a standard equation for a hyperbola with only right branch of the hyperbola. For small positive values of t, the path lies in the fourth quadrant and rises into the first quadrant as t increases, crossing the x-axis when $t = 1$. The parameter domain is $(0, \infty)$ and there is no starting point or terminal point for the path.

Remark: Any function can be rewritten parametrically in many different ways.

Example 8: In the case of Example 7, we can also represent the right-hand branch of the hyperbola by the parametrization $x = \sqrt{4 + t^2}, \quad y = t, \quad t \in \mathbb{R}$.
which is obtained by solving Equation (3) in Example 7 for $x \geq 0$ and letting y be the parameter t.
Another parametrization for the right-hand branch of this hyperbola given by Equation (3) is $x = 2\sec t, \quad y = 2\tan t, \quad -\pi/2 < t < \pi/2$. This parametrization follows from the trigonometric identity $\sec^2 t - \tan^2 t = 1$.
As t runs between $-\pi/2$ and $\pi/2$, $x = \sec t$ is positive and $y = \tan t$, runs between $-\infty$ and ∞, so P traverses the hyperbola's right-hand branch. It comes in along the branch's lower half as $t \to 0^-$, reaches $(2, 0)$ at $t = 0$, and moves out into the first quadrant as t increases steadily toward $\pi/2$. This is the same branch of the hyperbola shown Example 7.

10.2 Calculus with Parametric Curves

Tangent Lines and Areas

A parametrized curve $x = f(t)$ and $y = g(t)$ is **differentiable** at t if f and g are each differentiable at t. At a point on a differentiable parametrized curve where y is also a differentiable function of x, the derivatives dy/dt, dx/dt, and dy/dx are related by the Chain Rule:

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}.$$

If $dx/dt \neq 0$, we may divide both sides of this equation by dx/dt to solve for dy/dx.

Parametric Formula for dy/dx

If all three derivatives exist and $dx/dt \neq 0$, then $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$.

Parametric Formula for d^2y/dx^2

If the equations $x = f(t)$ and $y = g(t)$ define y as a twice-differentiable function of x, then at any point where $dx/dt \neq 0$ and $y' = dy/dx$, $\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt}$.

Example 1: Find the tangent line to the curve

$$x = \sec t, \quad y = \tan t, \quad -\pi/2 < t < \pi/2,$$

at the point $(\sqrt{2}, 1)$, where $t = \pi/4$ (Figure 1).

Solution:

The slope of the curve at t is $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\sec^2 t}{\sec t \tan t} = \frac{\sec t}{\tan t}$.

Setting t equal to $\pi/4$ gives

$$\left.\frac{dy}{dx}\right]_{t=\pi/4} = \frac{\sec(\pi/4)}{\tan(\pi/4)} = \frac{\sqrt{2}}{1} = \sqrt{2}.$$

The tangent line is $y - 1 = \sqrt{2}\,(x - \sqrt{2}\,)$, that is $y = \sqrt{2}\,x - 1$.

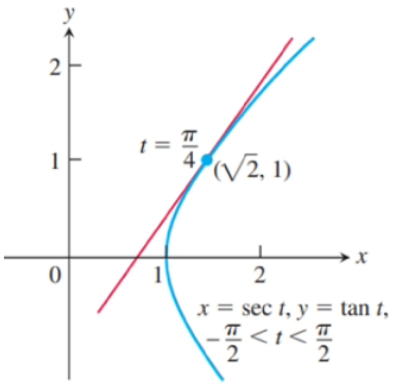


Figure 1: The curve is the right-hand branch of the hyperbola $x^2 - y^2 = 1$.

Finding d^2y/dx^2 in Terms of t

1. Express $y' = dy/dx$ in terms of t.
2. Find dy'/dt.
3. Divide dy'/dt by dx/dt.

Example 2: Find d^2y/dx^2 as a function of t if $x = t - t^2$ and $y = t - t^3$.

Solution:

$$y' = \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{1-3t^2}{1-2t}.$$

$$\frac{dy'}{dt} = \frac{d}{dt}\left(\frac{1-3t^2}{1-2t}\right) = \frac{2-6t+6t^2}{(1-2t)^2}.$$

$$\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt} = \frac{\frac{2-6t+6t^2}{(1-2t)^2}}{1-2t} = \frac{2-6t+6t^2}{(1-2t)^3}.$$

Example 3: Find the area enclosed by the astroid (Figure 2)

$$x = \cos^3 t, \quad y = \sin^3 t, \quad 0 \le t \le 2\pi.$$

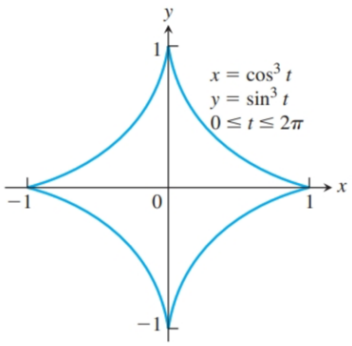


Figure 2: The astroid

Solution:
By symmetry, the enclosed area is four times the area beneath the curve in the first quadrant where $0 \le t \le \pi/2$.

$$A = 4\int_0^1 y\,dx$$

$$= 4\int_0^{\pi/2} (\sin^3 t)(3\cos^2 t \sin t)\,dt$$

$$= 12\int_0^{\pi/2} \left(\frac{1-\cos 2t}{2}\right)^2 \left(\frac{1+\cos 2t}{2}\right) dt$$

$$= \frac{3}{2}\int_0^{\pi/2} (1 - 2\cos 2t + \cos^2 2t)(1 + \cos 2t)\,dt$$

$$= \frac{3}{2}\int_0^{\pi/2} (1 - \cos 2t - \cos^2 2t + \cos^3 2t)\,dt$$

$$= \frac{3}{2}\left[\int_0^{\pi/2} (1-\cos 2t)\,dt - \int_0^{\pi/2} \cos^2 2t\,dt + \int_0^{\pi/2} \cos^3 2t\,dt\right]$$

$$= \frac{3}{2}\left[\left(t - \frac{1}{2}\sin 2t\right) - \frac{1}{2}\left(t + \frac{1}{4}\sin 2t\right) + \frac{1}{2}\left(\sin 2t - \frac{1}{3}\sin^3 2t\right)\right]_0^{\pi/2}$$

$$= \frac{3}{2}\left[\left(\frac{\pi}{2} - 0 - 0 - 0\right) - \frac{1}{2}\left(\frac{\pi}{2} + 0 - 0 - 0\right) + \frac{1}{2}(0 - 0 - 0 + 0)\right]$$

$$= \frac{3\pi}{8}.$$

Length of a Parametrically Defined Curve

Let C be a curve given parametrically by the equations

$$x = f(t) \quad \text{and} \quad y = g(t), \quad a \le t \le b$$

is a **smooth curve** if f and g are **continuously differentiable** (meaning they have continuous first derivatives) on the interval $[a, b]$, and the derivatives $f'(t)$ and $g'(t)$ are not simultaneously zero.

Definition 1: Length of Smooth Curve

If C is a smooth curve defined by $x = f(t)$ and $y = g(t)$, $a \le t \le b$, and C is traversed exactly once at t increases from $t = a$ to $t = b$, then the **length of** C is the definite integral

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2}\,dt.$$

If $x = f(t)$ and $y = g(t)$, then using the Leibniz notation we can write the formula for arc length this way:

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt.$$

Example 4: Find the length of the circle of radius r defined parametrically by

$$x = r\cos t \quad \text{and} \quad x = r\sin t, \quad 0 \le t \le 2\pi.$$

Solution :

As t varies from 0 to 2π, the circle is traversed exactly once, so the circumference is

$$L = \int_0^{2\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt.$$

$$\frac{dx}{dt} = -r\sin t \ , \quad \frac{dy}{dt} = r\cos t$$

and

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = (-r\sin t)^2 + (r\cos t)^2 = r^2.$$

Therefore, the total arc length is

$$L = \int_0^{2\pi} \sqrt{r^2}\, dt = r[t]_0^{2\pi} = 2\pi r.$$

Example 5: Find the length of the astroid (Figure 2)

$$x = \cos^3 t, \quad y = \sin^3 t, \quad 0 \le t \le 2\pi.$$

Solution:

Because of the curve's symmetry with respect to the coordinate axes, its length is four times the length of the first-quadrant portion.

$$\left(\frac{dx}{dt}\right)^2 = (-3\sin t\cos^2 t)^2 = 9\sin^2 t\cos^4 t\,,$$

$$\left(\frac{dy}{dt}\right)^2 = (3\cos t\sin^2 t)^2 = 9\cos^2 t\sin^4 t$$

and

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{9\sin^2 t\cos^2 t(\cos^2 t + \sin^2 t)^2}$$

$$= \sqrt{9\sin^2 t\cos^2 t} = 3|\sin t\cos t|$$

$$= 3\sin t\cos t \qquad \because \ \sin t\cos t \ge 0 \text{ for } 0 \le t \le \pi/2.$$

Therefore, Length of first-quadrant portion is

$$\int_0^{\pi/2} 3\sin t\cos t\, dt = \frac{3}{2}\int_0^{\pi/2} \sin 2t\, dt = -\frac{3}{4}[\cos 2t]_0^{\pi/4} = \frac{3}{2}.$$

The length of the astroid is four times this: 4(3/2) = 6.